

QE Without RE

Balance-Sheet Interventions and Expectational Coordination under Imperfect Knowledge

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Abstract

Balance-sheet policy is now a recurring part of monetary-policy implementation, but its expectational robustness is usually evaluated under rational expectations rather than under explicit belief formation. I study this distinction in a New Keynesian model with market segmentation, financial frictions, trend inflation, long-duration bonds, and adaptive learning. Asset-purchase rules that respond only to inflation or real activity have little effect on uniqueness or belief convergence. By contrast, rules that respond directly to bond prices can leave uniqueness largely intact while sharply reducing the region in which beliefs converge to the intended equilibrium. Within the class of rules studied here, preserving belief convergence then requires a stronger short-rate response to inflation. This effect is amplified when long-bond exposure is larger or when the purchased bond has longer duration. At the zero lower bound, the deflationary steady state remains locally unstable under learning, so belief perturbations are not pulled back to the low-rate equilibrium. In the face of shocks, balance-sheet policy can stabilize the real economy conditional on anchored beliefs, but can itself destabilize belief dynamics.

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I. INTRODUCTION

Balance-sheet policy has become a recurring component of monetary-policy implementation. Central banks now adjust not only short-term interest rates but also the size and composition of their asset holdings. The component studied here is the long-bond purchase-and-runoff part of balance-sheet policy.² Once balance-sheet policy is treated as a systematic policy instrument, the relevant question is not only how it affects output, inflation, and asset prices, but also whether the feedback it creates with private expectations preserves coordination on the intended equilibrium.

Because this long-bond purchase component operates through expectations-sensitive channels such as expected long-bond prices, its stabilization properties depend not only on equilibrium allocations but also on how beliefs are formed. Once policy transmission depends on beliefs, rational expectations (RE) is an incomplete benchmark: it characterizes outcomes under model-consistent forecasts, but is largely silent about whether agents converge to that equilibrium from nearby forecasting rules. Adaptive learning addresses that question by treating the RE equilibrium as a fixed point of the learning dynamics and by using expectational stability (E-stability, or learnability), that is, the local stability of a rational-expectations equilibrium under adaptive learning, to assess whether that fixed point is robust to belief errors. This distinction matters because determinacy and learnability are different objects. Determinacy asks whether equilibrium is unique under RE. E-stability asks whether recursive forecasting converges back to the RE fixed point from nearby beliefs. The two need not coincide, so a policy rule that delivers a unique RE equilibrium may still fail to support coordination under learning. Bullard and Mitra (2002) treat learnability as a necessary additional criterion for evaluating monetary policy rules, while Orphanides and Williams (2006) show that rules calibrated to perform well under RE may perform poorly when agents learn, so policy evaluation must be robust to imperfect knowledge. This paper therefore asks whether asset-purchase rules support stabilization once learnability is imposed, that is, whether they help agents coordinate on the intended equilibrium around the target-inflation steady state and at

²Throughout, QE denotes the long-bond purchase-and-runoff component of balance-sheet policy. In the model, this is the central bank's long-bond portfolio position.

the zero lower bound (ZLB).

To study this question, this paper starts from the four-equation dynamic stochastic general equilibrium model of Sims, Wu, and Zhang (2023), which features long-term bonds, financial intermediation, and real effects of balance-sheet interventions through market segmentation. I relax their full state-contingent inter-household bailout by keeping the steady-state bailout while removing the time-varying valuation component. Under this partial-bailout closure, fluctuations in the market value of long-term debt are no longer eliminated by construction, so bond prices remain part of the dynamic system. To model subjective expectations without breaking the model's intertemporal structure, I follow Woodford's (2013) temporary-equilibrium formulation, in which households and firms continue to solve infinite-horizon problems under subjective beliefs, while aggregate demand and supply can be written recursively in terms of composite expectational variables. This preserves intertemporal budget discipline under subjective expectations while keeping the forecasting problem one-step-ahead. I then replace rational expectations with adaptive learning as in Evans and Honkapohja (2001) and introduce trend inflation, following Ascari and Sbordone (2014). This serves two purposes. First, once the policy rule is constrained by the zero lower bound (ZLB), the model under RE admits not only the intended target-inflation steady state but also a deflationary steady state (Benhabib et al., 2001). Second, because E-stability is a local property, it must be evaluated using the coefficient matrices of the linearized system at the steady state under study. Trend inflation therefore matters directly for the exercise: it changes those local coefficients, and hence the Jacobian of the T-map on which E-stability is judged.

Several results emerge. First, asset-purchase rules that respond only to inflation or real activity have little effect on the rational-expectations determinacy frontier and only modest effects on the E-stability frontier under the benchmark calibration. The main stability effect arises when asset purchases respond directly to bond prices: the determinacy frontier moves little, but the E-stable region can shrink substantially. Thus a sizable determinate-but-E-unstable wedge can appear even when the rational-expectations equilibrium remains unique.

Second, this contraction has a policy-relevant implication within the class of rules studied here. Once balance-sheet policy responds to bond prices, the benchmark short-

rate calibration $(\phi_\pi^r, \phi_y^r) = (1.5, 0.125)$ lies closer to the boundary of the E-stable region. Preserving belief convergence then requires a stronger inflation response in the short-rate rule. The required response rises with the strength of bond-price-responsive asset purchases and is larger when long-bond exposure or bond duration is higher.

Third, at the ZLB deflationary steady state, the low-rate equilibrium remains locally unstable under learning. This does not mean that non-convergence is necessarily undesirable: belief perturbations may move the economy away from the deflationary steady state rather than back toward it. The point is instead that the low-rate equilibrium is not a robust anchor for expectations. The paper's contribution is therefore to show that, once belief formation is modeled explicitly, balance-sheet policy should be evaluated not only by its effects on rational-expectations determinacy and shock propagation, but also by whether the joint policy rule supports coordination on the intended equilibrium.

II. POSITION IN THE LITERATURE

This paper sits at the intersection of three literatures: adaptive learning, balance-sheet policy in DSGE models with financial frictions, and monetary policy under trend inflation. Its contribution is to connect these strands in a setting where asset-purchase rules are systematic policy instruments and the central question is not only how balance-sheet policy transmits under RE, but whether the resulting equilibrium is robust to belief errors under adaptive learning.

The paper is most directly related to the adaptive-learning literature. Evans and Honkapohja (2001) provide the canonical framework in which an RE equilibrium is treated as a fixed point of recursive forecasting, and E-stability governs whether beliefs converge to that fixed point. Bullard and Mitra (2002) show that determinacy and learnability are distinct criteria for evaluating monetary policy rules, so uniqueness under RE need not imply convergence under learning. Preston (2006) shows that forecast-based instrument rules may generate divergent learning dynamics and that the Taylor principle is not sufficient to guarantee convergence. More broadly, Eusepi and Preston (2018) argue that imperfect knowledge can narrow the set of policies available to central banks and complicate expectations management. Relative to this literature,

the present paper adds a balance-sheet instrument and studies how asset-purchase feedback rules affect both determinacy and E-stability around the relevant steady states.

The paper also connects to the QE literature. Empirically, Gagnon et al. (2011) show that large-scale asset purchases reduce longer-term interest rates, primarily through lower risk premia rather than lower expected future short rates. Krishnamurthy and Vissing-Jorgensen (2011) show that QE works through multiple channels with asset-specific effects, which suggests transmission beyond a pure expectations-hypothesis view. Theoretically, Gertler and Karadi (2011) and Gertler and Karadi (2013) model unconventional policy as central-bank intermediation operating through limits to private arbitrage, while Sims and Wu (2021) studies QE, forward guidance, and negative interest rate policy in a richer DSGE environment. Sims et al. (2023) distill that class of mechanisms into a tractable four-equation New Keynesian model with financial intermediation, short- and long-term bonds, credit shocks, and scope for balance-sheet policy.

A smaller related literature combines balance-sheet policy with boundedly rational expectations. Iovino and Sergeyev (2023) study balance-sheet interventions under iterative level- k thinking and show that such interventions can be effective even when they are neutral under RE. They also show that interventions make forecast errors about future endogenous variables predictable. Cole and Huh (2024) compare forward guidance and QE under adaptive learning in Sims et al.'s (2023) environment and find that forward guidance has a more pronounced effect, especially on inflation, than QE. Relative to these papers, the present paper asks a different question: not whether departures from RE create a role for QE or whether QE is more powerful than other unconventional tools, but whether QE feedback rules preserve coordination on the intended equilibrium when agents learn.

Finally, the paper connects to the literature on trend inflation. Ascari and Ropele (2009) show that positive trend inflation shrinks the determinacy region and that the Taylor principle is not sufficient for local determinacy. Kurozumi (2014) extends the issue to adaptive learning by studying E-stability under trend inflation in a New Keynesian model. Relative to this work, the present paper studies how adding a balance-sheet instrument changes determinacy and learnability near both the

target-inflation and deflationary steady states.

Taken together, these literatures leave open an important question. Existing work has studied adaptive learning, QE transmission, and trend inflation largely separately. This paper brings them together and evaluates balance-sheet policy not only by its effects on output, inflation, and asset prices, but also by whether the joint short-rate and asset-purchase rule remains learnable.

III. MODEL

This paper derives the model using an agent-level approach. In the context of bounded rationality, this method provides microfoundations and robustness to the Lucas critique, and it can generate nontrivial differences because the market-clearing condition, an outcome of individual decisions after aggregation, is imposed only after bounded rationality has been introduced at the agent level (Evans and McGough, 2020). Let $\widehat{E}_t$ denote agents' subjective expectations conditional on information available at time t . When the subjective joint distribution of states coincides with its objective counterpart, the economy operates under fully rational expectations, denoted by E_t . In this paper, adaptive learning is the form through which bounded rationality is manifested. Expectations are also assumed to be homogeneous, although individual agents do not know this when making their decisions. Multi-step forecasts are formed by iterating the agents' time- t perceived law of motion forward. Thus $\widehat{E}_t$ denotes a forecast operator based on current beliefs, rather than a dynamically consistent family of conditional expectations under a fixed subjective probability measure.

III.A. ENVIRONMENT AND SETUP

The economy consists of a continuum of two types of infinitely lived households, indexed by $i \in [0, 1]$. There is also a continuum of production firms, indexed by $j \in [0, 1]$, and a representative financial intermediary, both of which are owned by patient households. The government consists of a central bank and a fiscal authority.

Patient households engage in consumption, saving, labor supply, and ownership of the financial intermediary and production firms. In contrast, impatient households

consume and borrow, do not supply labor, and receive transfers X_t^I from patient households to cover part of their debt. These roles are fixed throughout the economy, with $z \in [0, 1)$ denoting the share of impatient households. I assume that households live in a community with within-group risk sharing, so patient households share the aggregate transfer to impatient households equally each period, $\frac{1}{\int_{i=z}^1 di} X_t^I$. Consequently, each impatient household receives $\frac{1}{\int_{i=0}^z di} X_t^I$. The financial intermediary arbitrages between deposits, reserves, and long-term bonds, but is subject to a leverage constraint. The central bank controls the rate on reserves and its long-bond holdings, so balance-sheet policy enters the model as a second policy instrument alongside the short-term interest rate.

Throughout, bar variables denote deterministic steady states on the trend-inflation path, and inflation is written in stationary form as

$$\pi_t \equiv \log\left(\frac{\Pi_t}{\bar{\Pi}}\right), \quad (1)$$

where $\Pi_t \equiv P_t/P_{t-1}$ and $\bar{\Pi}$ is gross trend inflation. The distinction between steady states matters because trend inflation changes the local coefficient matrices around each steady state, which in turn changes the Jacobian relevant for E-stability. In the main text, hats are suppressed on first-order variables. The variables π_t , q_t , $r_{t-1,t}^s$, $r_{t-1,t}^b$, and qe_t are log deviations from their own steady-state values. The variables y_t , $c_{s,t}$, $c_{b,t}$, g_t , h_t , $h_{b,t}$, τ_t , d_t^{FI} , x_t^{FI} , s_t , and b_t are deviations of the corresponding real quantities from steady state divided by $\bar{Y}$, with the precise normalizations for s_t and b_t introduced below. Detailed derivations are given in the Appendix.

III.A.1. PATIENT HOUSEHOLDS

Patient households consume, save by purchasing one-period bonds issued by the financial intermediary, supply labor inelastically at the union-determined wage, and own the financial intermediary and production firms. I assume standard monotonicity, concavity, and Inada conditions for preferences. Each patient household i solves

$$\max_{\{C_{s,T}^i, S_T^i\}_{T=t}^{\infty}} \widehat{E}_t \sum_{T=t}^{\infty} \beta^{T-t} [u(C_{s,T}^i; \xi_T) - v(L_T^i; \xi_T)], \quad (2)$$

subject to

$$P_t C_{s,t}^i + S_t^i \leq R_{t-1,t}^s S_{t-1}^i + P_t H_t, \quad (3)$$

where $H_t \equiv (Y_t + T_t + D_t^{FI} - X_t^{FI} - X_t^I) / \int_{i=z}^1 di$ denotes patient-household disposable income. Here Y_t is aggregate output, T_t denotes government transfers, D_t^{FI} is the financial intermediary's dividend payment to patient households, X_t^{FI} is the transfer from patient households to the intermediary, and X_t^I is the inter-household transfer to impatient households. Following Woodford (2013), wages are set by a union that bargains collectively on behalf of households, so patient households supply labor as demanded by firms.

Let $S_t^i \equiv S_t^i / P_t$ denote the stationary real asset stock. In the linearized system, $s_t^i \equiv (S_t^i - \bar{s}) / \bar{Y}$, $h_t \equiv (H_t - \bar{H}) / \bar{Y}$, and $\chi_s \equiv \bar{s} / \bar{Y}$. Combining the Euler equation with the lifetime budget constraint under subjective beliefs gives the log-linearized patient-household temporary-equilibrium consumption rule

$$\begin{aligned} c_{s,t}^i = & \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) s_{t-1}^i + \beta \zeta_t + (1 - \beta) h_t + \chi_s \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) (r_{t-1,t}^s - \pi_t) \\ & - \left[\sigma - \chi_s \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) \right] \beta r_{t,t+1}^s + \beta \widehat{E}_t v_{s,t+1}^i, \end{aligned} \quad (4)$$

where the composite term $v_{s,t}^i$ collects household i 's higher-horizon forecast terms and satisfies

$$v_{s,t}^i = (1 - \beta)(h_t - \zeta_t) - \left[\sigma - \chi_s \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) \right] (\beta r_{t,t+1}^s - \pi_t) + \beta \widehat{E}_t v_{s,t+1}^i. \quad (5)$$

The derivation is given in Section A.1.

III.A.2. IMPATIENT HOUSEHOLDS

Impatient households consume, issue long-term bonds, and receive the inter-household transfer from patient households. They do not supply labor. Long bonds are perpetuities with coupon payments that decay geometrically at rate $\kappa \in [0, 1]$: one unit issued at time t pays one unit at $t + 1$, κ units at $t + 2$, and so on. The total liability

of debtor i is therefore

$$B_t^i = NB_t^i + \kappa NB_{t-1}^i + \kappa^2 NB_{t-2}^i + \cdots, \quad (6)$$

so net issuance is $NB_t^i = B_t^i - \kappa B_{t-1}^i$. Long bonds trade at price Q_t , and the gross nominal return is $R_{t-1,t}^b = (1 + \kappa Q_t)/Q_{t-1}$. Impatient household i solves

$$\max_{\{C_{b,T}^i, B_T^i\}_{T=t}^{\infty}} \widehat{E}_t \sum_{T=t}^{\infty} \beta_b^{T-t} u(C_{b,T}^i; \xi_T), \quad (7)$$

subject to

$$P_t C_{b,t}^i + B_{t-1}^i \leq Q_t (B_t^i - \kappa B_{t-1}^i) + P_t H_{b,t}, \quad (8)$$

where $H_{b,t} \equiv X_t^I / \int_{i=0}^z di$ is the transfer received by each impatient household. Let $\mathcal{B}_t^i \equiv B_t^i / P_t$ denote the stationary real liability stock. In steady state, $\bar{R}^b = \bar{\Pi} / \beta_b$. In the linearized system, $b_t^i \equiv \bar{Q}(\mathcal{B}_t^i - \bar{b}) / \bar{Y}$, $h_{b,t} \equiv (H_{b,t} - \bar{H}_b) / \bar{Y}$, and $\chi_b \equiv \bar{Q} \bar{b} / \bar{Y}$. Combining the borrower Euler equation with the lifetime budget constraint gives the log-linearized borrower temporary-equilibrium consumption rule

$$\begin{aligned} c_{b,t}^i = & - \left(\frac{\bar{R}^b}{\bar{\Pi}} - 1 \right) b_{t-1}^i + \beta_b \zeta_t \\ & + (1 - \beta_b) \left[\chi_b \left(1 - \frac{\kappa}{\bar{\Pi}} \right) q_t + \chi_b \frac{\bar{R}^b}{\bar{\Pi}} \pi_t + h_{b,t} \right] + \beta_b \widehat{E}_t v_{b,t+1}^i, \end{aligned} \quad (9)$$

with composite expectations term recursion

$$\begin{aligned} v_{b,t}^i = & (1 - \beta_b) \left[\chi_b \left(1 - \frac{\kappa}{\bar{\Pi}} \right) q_t + h_{b,t} - \zeta_t \right] - \sigma r_{t-1,t}^b \\ & + \left[\sigma + \chi_b \left(\frac{\bar{R}^b}{\bar{\Pi}} - 1 \right) \right] \pi_t + \beta_b \widehat{E}_t v_{b,t+1}^i, \end{aligned} \quad (10)$$

where $1 - \kappa / \bar{\Pi}$ is the trend-inflation correction implied by the stationarized long-bond budget constraint. The derivation is given in Section A.1.

A key simplifying assumption in Sims et al. (2023) is the full-bailout assumption,

$$P_t X_t^I = (1 + \kappa Q_t) \int_{i=0}^z B_{t-1}^i di.$$

This state-contingent inter-household transfer offsets the period- t market value of outstanding impatient-household liabilities, insulating borrowers from bond-price movements and making their problem static.

I instead use a partial-bailout closure. Patient households continue to provide the steady-state bailout to impatient households, but the transfer does not vary with the current market value of outstanding long-term liabilities. Formally, the real transfer is fixed at its steady-state value,

$$\bar{X}^I = \frac{1 + \kappa \bar{Q}}{\bar{\Pi}} \int_{i=0}^z \bar{b}^i di.$$

Thus the model does not eliminate the bailout margin, it removes only the time-varying, state-contingent valuation component of the full bailout. Under this partial-bailout closure, fluctuations in the market value of long-term debt are no longer absorbed period by period by inter-household transfers. Movements in the long-bond price therefore remain in the impatient household's budget constraint and become an endogenous component of the dynamic system. Since $X_t^I = \bar{X}^I$, each impatient household receives the same steady-state transfer every period, so the log-linearized impatient-household transfer term satisfies $h_{b,t} = 0$ everywhere.

III.A.3. FINANCIAL INTERMEDIARY AND GOVERNMENT

Financial markets are segmented: patient households hold short bonds issued by the intermediary, while impatient households issue long-term bonds held by the intermediary and the central bank. The representative intermediary, owned by patient households, raises funds from deposits and from the transfer X_t^{FI} , and allocates them between long-term bonds and reserves. Its balance-sheet condition is

$$Q_t B_t^{FI} + RE_t^{FI} = S_t^{FI} + P_t X_t^{FI}, \quad (11)$$

and its long-bond position is constrained by

$$Q_t B_t^{FI} \leq \Theta_t P_t \bar{X}^{FI}, \quad (12)$$

where Θ_t captures credit conditions and $\bar{X}^{FI}$ is steady-state intermediary net worth. Reserves receive zero weight in the constraint, so a decline in Θ_t reduces the intermediary's effective risk-bearing capacity for long bonds. At the end of each period, the intermediary returns the net proceeds from long bonds and reserves, net of interest paid on deposits, to patient households. In the reduced system, this income effect is summarized by the dividend d_t^{FI} and the transfer x_t^{FI} .

The central bank adjusts its long-bond portfolio through reserve creation,

$$Q_t B_t^{CB} = RE_t^{CB}, \quad (13)$$

and the consolidated public-sector budget is

$$P_t T_t = R_{t-1,t}^b Q_{t-1} B_{t-1}^{CB} - R_{t-1,t}^{re} RE_{t-1}^{CB} - P_t G_t. \quad (14)$$

The central bank controls the reserve rate $R_{t,t+1}^{re}$. Since reserves and short-term deposits are perfect substitutes for the intermediary, $r_{t,t+1}^{re} = r_{t,t+1}^s$. Conventional policy therefore operates through the short rate, while balance-sheet policy operates through the central bank's long-bond position.

III.A.4. MARKET CLEARING

Goods-market clearing and segmented asset-market clearing require

$$Y_t = \int_{i=0}^z C_{b,t}^i di + \int_{i=z}^1 C_{s,t}^i di + G_t, \quad (15)$$

$$\int_{i=z}^1 S_t^i di = S_t^{FI}, \quad (16)$$

$$\int_{i=0}^z B_t^i di = B_t^{FI} + B_t^{CB}, \quad (17)$$

$$RE_t^{FI} = RE_t^{CB}. \quad (18)$$

With homogeneous expectations and within-household-type symmetry, households of a given type make identical decisions in equilibrium, so I drop the individual index i after aggregation and write type-level allocations as $c_{s,t}$, $c_{b,t}$, $v_{s,t}$, $v_{b,t}$, s_t , and b_t . Together with Equations (11) to (13), these conditions determine the equilibrium links among deposits, long-bond holdings, reserves, and central-bank asset purchases.

III.B. MODEL IN LOG-LINEARIZED FORM

III.B.1. AGGREGATE DEMAND BLOCK

Combining Equations (4), (5), (9) and (10) with goods-market clearing in Equation (15) yields the aggregate-demand curve

$$y_t = (1 - z) \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) s_{t-1} - z \left(\frac{\bar{R}^b}{\bar{\Pi}} - 1 \right) b_{t-1} + z\sigma(r_{t-1,t}^b - \pi_t) \\ + (1 - z)\sigma \left[\frac{\chi_s}{\sigma} \left(\frac{\bar{R}^s}{\bar{\Pi}} - 1 \right) r_{t-1,t}^s - \pi_t \right] + zv_{b,t} + (1 - z)v_{s,t} + g_t + \zeta_t, \quad (19)$$

The lagged asset positions s_{t-1} and b_{t-1} remain in Equation (19) even though deposits and long bonds clear in equilibrium. The reason is that Equation (19) is derived from household demand equations, which depend on beginning-of-period positions carried into period t . Market clearing determines the counterpart holders of those positions, namely the intermediary for deposits and the intermediary together with the central bank for long bonds, but it does not remove their effect on current household consumption. Because markets are segmented, these positions cannot be netted out within the household sector at the aggregation stage. They cancel only in the fully consolidated economy. Hence s_{t-1} and b_{t-1} remain in Equation (19).

Together with forward recursions Equation (5) and Equation (10) for $v_{s,t}$ and $v_{b,t}$, and borrower's consumption path Equation (9), these equations form the aggregate demand block of the model.

III.B.2. NKPC BLOCK UNDER TREND INFLATION

On the supply side, trend inflation and indexation replace the standard forward-looking NKPC with a three-equation block written in quasi-differenced inflation. Production

uses labor as the sole input, with capital normalized to one, so $Y_t = A_t L_t^{1-\alpha_y}$. Define

$$\tilde{\pi}_t \equiv \pi_t - \gamma_\pi \pi_{t-1}, \quad (20)$$

where $\gamma_\pi \in [0, 1]$ is the indexation parameter. The pricing-kernel and demand objects are

$$k_t^p \equiv v\varpi_t + \frac{\nu+1}{1-\alpha_y}(y_t - a_t) - v\eta_t, \quad u_t^d \equiv y_t - \frac{1}{\sigma(1-z)}(y_t - zc_{b,t} - g_t) + \frac{1}{\sigma}\zeta_t, \quad (21)$$

where ϖ_t is price dispersion, a_t is technology, and η_t is the markup disturbance. The generalized NKPC block is

$$\varpi_t = \delta_\omega \varpi_{t-1} + \omega_\Delta \tilde{\pi}_t, \quad (22)$$

$$\psi_t = (1 - \delta_n)k_t^p + \delta_n \widehat{E}_t(\psi_{t+1} + \mathcal{K}_p \tilde{\pi}_{t+1}), \quad (23)$$

$$\begin{aligned} \chi_\Delta \tilde{\pi}_t &= (1 - \delta_n)k_t^p - (1 - \delta_d)u_t^d + (\delta_n - \delta_d)\widehat{E}_t\psi_{t+1} \\ &\quad + [\delta_n \mathcal{K}_p - \delta_d((\theta_p - 1) - \chi_\Delta)] \widehat{E}_t \tilde{\pi}_{t+1}. \end{aligned} \quad (24)$$

Here ψ_t is the auxiliary forward-looking object in firms' reset-price problem, α_p is the Calvo non-adjustment probability, θ_p is the elasticity of substitution across goods, $\Gamma_\Pi \equiv \bar{\Pi}^{(\theta_p-1)(1-\gamma_\pi)}$, $\mathcal{K}_p \equiv \theta_p/(1-\alpha_y)$, $\mathcal{K}_a \equiv 1 + \theta_p \alpha_y/(1-\alpha_y)$, $\delta_d \equiv \alpha_p \beta \Gamma_\Pi$, $\delta_n \equiv \alpha_p \beta \bar{\Pi}^{\mathcal{K}_p(1-\gamma_\pi)}$, $\delta_\omega \equiv \alpha_p \bar{\Pi}^{\mathcal{K}_p(1-\gamma_\pi)}$, $\chi_\Delta \equiv \mathcal{K}_a \alpha_p \Gamma_\Pi/(1 - \alpha_p \Gamma_\Pi)$, and $\omega_\Delta \equiv \delta_\omega \mathcal{K}_p - \mathcal{K}_p(1 - \delta_\omega) \alpha_p \Gamma_\Pi/(1 - \alpha_p \Gamma_\Pi)$.

III.B.3. BOND PRICING AND BALANCE-SHEET RELATIONS

The remaining equations describe long-bond pricing, the intermediary balance-sheet wedge, portfolio clearing, and the transfer terms that enter patient-household income. Long-bond pricing and the corresponding realized return are

$$q_t = \frac{1}{\sigma}(c_{b,t} - \zeta_t) - \frac{1}{\sigma}\widehat{E}_t(c_{b,t+1} - \zeta_{t+1}) - \widehat{E}_t\pi_{t+1} + \frac{\kappa\beta_b}{\bar{\Pi}}\widehat{E}_t q_{t+1}, \quad (25)$$

$$r_{t-1,t}^b = \frac{\kappa\beta_b}{\bar{\Pi}}q_t - q_{t-1}. \quad (26)$$

The intermediary balance-sheet wedge is

$$x_t^{FI} = \frac{\kappa}{\bar{\Pi}} \chi_f (q_t - q_{t-1} - \pi_t + \theta_{t-1}), \quad (27)$$

where $\chi_f \equiv \bar{Q}\bar{b}^{FI}/\bar{Y}$ is the intermediary's steady-state long-bond position relative to output and θ_t is the credit disturbance. The leverage constraint, intermediary balance sheet, central-bank balance sheet, and segmented asset-market clearing imply

$$b_t = \frac{\chi_c q e_t + \chi_f \theta_t - z \chi_b q_t}{z}, \quad (28)$$

$$s_t = \frac{\chi_c q e_t + \chi_f \theta_t - x_t^{FI}}{1 - z}, \quad (29)$$

where $\chi_c \equiv \bar{Q}\bar{b}^{CB}/\bar{Y}$. The variable $q e_t$ denotes the first-order deviation of the real market value of the central bank's long-bond portfolio, $Q_t B_t^{CB}/P_t$, from steady state, scaled so that $\chi_c q e_t$ is the output-scaled market-value deviation. With this convention, the corresponding physical-holdings deviation valued at the steady-state bond price is $\chi_c (q e_t - q_t)$. Portfolio clearing then implies $z \chi_b = \chi_c + \chi_f$.

The fiscal-transfer and intermediary-dividend terms entering patient-household income are

$$\tau_t = \chi_c \left(\frac{\bar{R}^b}{\bar{\Pi}} r_{t-1,t}^b - \frac{\bar{R}^s}{\bar{\Pi}} r_{t-1,t}^s \right) + \chi_c \left(\frac{\bar{R}^b - \bar{R}^s}{\bar{\Pi}} \right) q_{t-1} - \chi_c \left(\frac{\bar{R}^b - \bar{R}^s}{\bar{\Pi}} \right) \pi_t - g_t, \quad (30)$$

$$\begin{aligned} d_t^{FI} = & \chi_f \left(\frac{\bar{R}^b}{\bar{\Pi}} r_{t-1,t}^b - \frac{\bar{R}^s}{\bar{\Pi}} r_{t-1,t}^s \right) + \chi_f \left(\frac{\bar{R}^b - \bar{R}^s}{\bar{\Pi}} \right) \theta_{t-1} - \chi_f \left(\frac{\bar{R}^b - \bar{R}^s}{\bar{\Pi}} \right) \pi_t \\ & + \frac{\bar{R}^s}{\bar{\Pi}} \left[\frac{\tilde{X}^{FI}}{\bar{Y}} (r_{t-1,t}^s - \pi_t) + x_{t-1}^{FI} \right], \end{aligned} \quad (31)$$

where $\tilde{X}^{FI}$ denotes the intermediary's steady-state transfer from patient households and satisfies $\tilde{X}^{FI}/\bar{Y} = z \chi_b - (1 - z) \chi_s$. These transfer terms enter the system through h_t in Equation (5).

III.B.4. POLICY RULES

Under contemporaneous policy rules, the central bank sets the reserve rate and its long-bond position according to

$$r_{t,t+1}^{re} = \phi_{\pi}^r \pi_t + \phi_y^r y_t + \phi_q^r q_t, \quad (32)$$

$$qe_t = -(\phi_{\pi}^{qe} \pi_t + \phi_y^{qe} y_t + \phi_q^{qe} q_t). \quad (33)$$

The central bank adjusts QE in the opposite direction from the short-term interest rate. The negative sign in the QE rule means that the balance sheet expands when the target variable falls below steady state. In particular, when $\phi_q^{qe} > 0$, a decline in the long-bond price raises qe_t , so bond-price-responsive QE leans against falling bond prices. Since reserves and short bonds are perfect substitutes for the intermediary, the reserve rate transmits one for one to the short rate, $r_{t,t+1}^{re} = r_{t,t+1}^s$. From this point onward I write $r_t \equiv r_{t,t+1}^s = r_{t,t+1}^{re}$.

III.C. PARAMETER CALIBRATION

IV. DETERMINACY & E-STABILITY

Determinacy and E-stability are characterized numerically under the calibration in Table 1. The two concepts require different state-space representations. Determinacy is a property of the rational-expectations equilibrium system, so it is invariant to alternative companion representations that preserve the same equilibrium conditions. E-stability, by contrast, is a property of the learning map from the perceived law of motion (PLM) to the actual law of motion (ALM), so it depends on the information timing and state variables that agents use in forecasting. For this reason, determinacy is studied using a companion-form representation convenient for QZ decomposition, whereas E-stability is studied using the minimal state variable (MSV) solution PLM.

IV.A. DETERMINACY

For determinacy, I abstract from exogenous processes and write the RE system in companion form, since stationary exogenous processes do not affect the local

Table 1: Benchmark Calibration

Symbol	Value	Description	Target / Interpretation
<i>Households and portfolio structure</i>			
β	0.995	Patient-household discount factor	Quarterly value.
β_b	0.990	Impatient-household discount factor	
z	1/3	Share of impatient households	
σ	1	Intertemporal elasticity of substitution	Coupon stream with 40-quarter duration.
κ	$1 - 40^{-1}$	Coupon-decay parameter of long bond	
χ_s	1/5	Steady-state patient short-asset position relative to output	
χ_b	4/5	Steady-state impatient long-bond liability relative to output	Stock-flow normalization for patient-household deposits, implies aggregate short claims $(1 - z)\chi_s \approx 0.133\bar{Y}$, keeping saver balance-sheet exposure positive but secondary to long-bond exposure. Chosen jointly with z so aggregate long-bond exposure is $z\chi_b \approx 0.267\bar{Y}$.
χ_c	$\frac{1}{3}z\chi_b$	Central-bank long-bond position relative to output	Sets central-bank long-bond holdings to about $0.089\bar{Y}$, close to the 0.1 $\bar{Y}$ steady-state QE target in Sims et al. (2023).
χ_f	$\frac{2}{3}z\chi_b$	Intermediary long-bond position relative to output	Close to the 70/30 intermediary/central-bank split in Sims et al. (2023).
<i>Trend inflation and price setting</i>			
$\bar{\Pi}$		Quarterly gross steady-state inflation	$1.02^{1/4}$ (0.98 $^{1/4}$) for target-inflation (deflationary) steady state.
α_p	0.75	Calvo non-adjustment probability	Average price duration of four quarters.
θ_p	6	Elasticity of substitution across goods	Implies a 20% steady-state markup.
γ_π	0	Inflation indexation parameter	No indexation in the baseline.
α_y	0	Decreasing-returns parameter	Constant returns-to-scale benchmark.
ν	1	Pricing-kernel curvature	
<i>Shock processes</i>			
$\rho_\zeta = \rho_\eta =$	0.8	Common shock persistence	AR(1) persistence for preference, markup, government-spending, technology, and credit shocks.
$\rho_g = \rho_a = \rho_\theta$			

equilibrium classification,

$$U_t \equiv \begin{bmatrix} q_{t-2} & q_{t-1} & \pi_{t-1} & r_{t-1} & qe_{t-1} & \varpi_{t-1} & q_t & \pi_t & y_t & v_{b,t} & v_{s,t} & c_{b,t} & \varpi_t & \psi_t & r_t & qe_t \end{bmatrix}'. \quad (34)$$

The system can then be written as

$$A_C E_t U_{t+1} = B_C U_t. \quad (35)$$

The predetermined variables are the lagged objects q_{t-2} , q_{t-1} , π_{t-1} , r_{t-1} , qe_{t-1} , and ϖ_{t-1} , so the contemporaneous system has six state variables. The second lag q_{t-2} enters because lagged intermediary-transfer terms contain x_{t-1}^{FI} , which depends on q_{t-2} .

I compute the generalized eigenvalues of the pencil (A_C, B_C) using the QZ method in Klein (2000) and classify the equilibrium following Blanchard and Kahn (1980). The equilibrium is determinate when the number of generalized eigenvalues inside the unit circle equals the number of predetermined variables. If that number is smaller, the system is explosive. If it is larger, the model is indeterminate.

IV.B. MSV SOLUTIONS, T-MAP, AND E-STABILITY

For learning, define the endogenous vector

$$Z_t \equiv \begin{bmatrix} q_t & \pi_t & y_t & v_{b,t} & v_{s,t} & c_{b,t} & \varpi_t & \psi_t & r_t & qe_t \end{bmatrix}'. \quad (36)$$

The MSV state consists of the predetermined endogenous lags that matter for forecasting together with the augmented exogenous block,

$$\ell_{t-1} \equiv \begin{bmatrix} q_{t-1} & q_{t-2} & \pi_{t-1} & r_{t-1} & qe_{t-1} & \varpi_{t-1} \end{bmatrix}', \quad (37)$$

$$n_t \equiv \begin{bmatrix} \zeta_t & \eta_t & g_t & a_t & \theta_t & \theta_{t-1} & \theta_{t-2} \end{bmatrix}', \quad w_t \equiv \begin{bmatrix} \ell_{t-1} \\ n_t \end{bmatrix}. \quad (38)$$

Agents are assumed to estimate the MSV PLM

$$Z_t = \mu_Z + \Phi_\ell \ell_{t-1} + \Phi_n n_t = \mu_Z + \Phi w_t, \quad \Phi \equiv \begin{bmatrix} \Phi_\ell & \Phi_n \end{bmatrix}. \quad (39)$$

Here Z_t is 10×1 , ℓ_{t-1} is 6×1 , n_t is 7×1 , and w_t is 13×1 . Thus Φ_ℓ is 10×6 , Φ_n is 10×7 , and Φ is 10×13 . Current exogenous states in n_t are observed when agents form $\widehat{E}_t$ -forecasts. The state then evolves according to

$$w_{t+1} = A_w w_t + B_z Z_t + \tilde{\varepsilon}_{t+1}. \quad (40)$$

Here $\tilde{\varepsilon}_{t+1}$ denotes the innovation vector in the stacked state-space system: it is zero in the endogenous-lag block and contains the shock innovation in the augmented exogenous block.

Under contemporaneous policy rules, the reduced system can be written as

$$Z_t = \mathcal{F}_2 \widehat{E}_t Z_{t+2} + \mathcal{F}_1 \widehat{E}_t Z_{t+1} + \mathcal{F}_w w_t. \quad (41)$$

The matrices $\mathcal{F}_2$ and $\mathcal{F}_1$ are 10×10 , and $\mathcal{F}_w$ is 10×13 . Appendix C gives the corresponding scalar-to-matrix construction. Substituting the PLM-inferred $\widehat{E}_t Z_{t+1}$ and $\widehat{E}_t Z_{t+2}$ into Equation (41) yields the ALM

$$Z_t = \mathcal{T}_0(\mu_Z, \Phi) + \mathcal{T}_w(\mu_Z, \Phi) w_t. \quad (42)$$

This defines the MSV T-map, which maps PLM coefficients into ALM coefficients,

$$\mathcal{T}^{msv}(\mu_Z, \Phi) = (\mathcal{T}_0(\mu_Z, \Phi), \mathcal{T}_w(\mu_Z, \Phi)). \quad (43)$$

To study E-stability, stack the perceived coefficients as $\xi \equiv [\mu'_Z, \text{vec}(\Phi)']'$. Under the Evans and Honkapohja (2001) learning ODE

$$\frac{d\xi}{d\tau} = \mathcal{T}^{msv}(\xi) - \xi, \quad (44)$$

a fixed point ξ^* is locally E-stable if

$$\Re \lambda(D\mathcal{T}^{msv}(\xi^*)) < 1, \quad (45)$$

that is, all eigenvalues of the Jacobian must have real part below one. Appendix C describes the numerical solution and classification procedure.

V. RESULTS

V.A. TARGET-INFLATION STEADY STATE: STABILIZATION VERSUS EXPECTATIONAL FRAGILITY

I begin at the target-inflation steady state, with annualized trend inflation fixed at 2%, so that $\bar{\Pi} = 1.02^{1/4}$, under the contemporaneous benchmark short-rate rule $(\phi_\pi^r, \phi_y^r) = (1.5, 0.125)$. Figure 1 reports determinacy and E-stability regions for four balance-sheet-policy specifications. Rules that respond only to output or inflation leave both frontiers close to the QE-off benchmark. By contrast, when asset purchases respond directly to bond prices, the determinacy frontier moves only modestly but the E-stable region contracts substantially. A determinate-but-E-unstable wedge therefore appears even though the rational-expectations equilibrium remains unique. The benchmark point “x” remains on the E-stable side of the boundary, so local learning impulse responses around this point remain meaningful.

I next ask how the same policy rule affects transition dynamics conditional on remaining near the intended equilibrium. The learning impulse responses are simulation-based local generalized impulse responses under recursive least-squares adaptive learning. Beliefs are initialized from the RE solution mapped into the MSV PLM, then updated through a pre-sample learning simulation. For each replication, I simulate two economies with identical background shocks, except that one receives an additional credit shock at impact. Both economies update beliefs during the response horizon using the same constant-gain recursive least-squares rule. The impulse-response simulation uses delayed-information updating commonly used in the literature, so the regression update after observing period- t outcomes uses the

Contemporaneous rule: target-inflation steady state

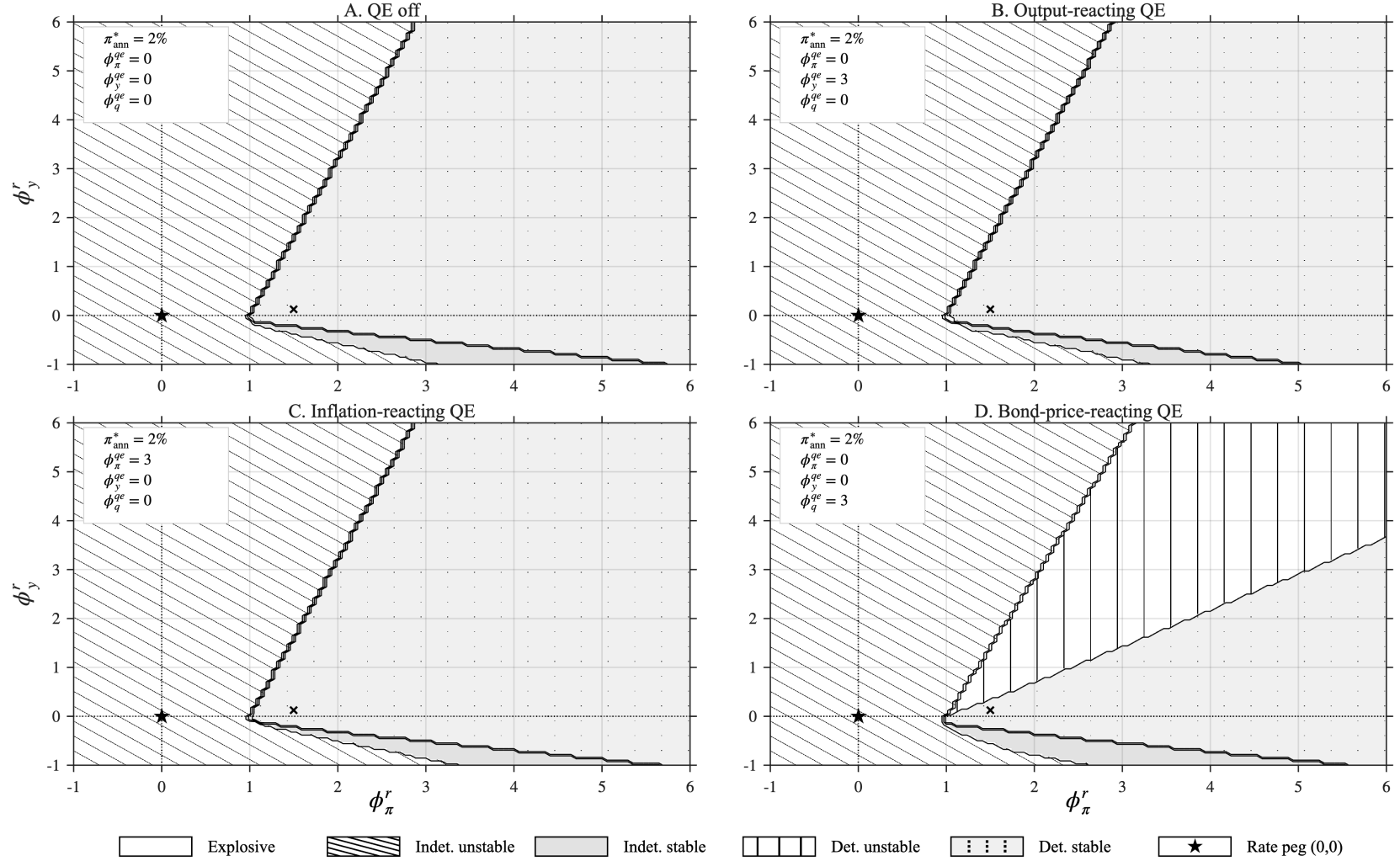


Figure 1: Determinacy and E-stability regions under the contemporaneous rule at the target-inflation steady state. The marker “x” indicates the benchmark policy point $(\phi_\pi^r, \phi_y^r) = (1.5, 0.125)$.

previously available state vector rather than the contemporaneous shock-augmented state. The reported learning response is the average difference between the shocked and unshocked paths.

Figure 3 reports impulse responses to a negative credit shock, that is, a tightening in intermediary risk-bearing capacity. The figure compares QE off with a bond-price-responsive rule, $qe_t = -\phi_q^{qe} q_t$, setting $\phi_q^{qe} = 3$. Without QE, the shock lowers the long-bond price, raises expected bond returns and excess returns, contracts private asset positions, and pushes output and inflation below steady state. Bond-price-responsive QE attenuates these responses under rational expectations. Under learning, the impact responses are not uniformly smaller at every horizon, but the subsequent overshooting of output, inflation, and bond prices is reduced. In this sense, QE is conditionally stabilizing: given convergence to the intended equilibrium, it leans against the financial amplification triggered by the credit shock.

The conditional stabilization in Figure 3 and the inward movement of the E-stability boundary in Figure 1 come from the same feedback channel. A fall in q_t triggers bond purchases, so the policy rule leans directly against the financial margin hit by the credit shock. But the rule also makes realized outcomes more sensitive to beliefs about q_t , the asset price to which policy responds. The remaining target-inflation exercises quantify this local trade-off and isolate the mechanism.

For a given balance-sheet rule, define the restoration threshold as the smallest short-rate inflation coefficient that restores local E-stability,

$$\phi_{\pi}^{r,*}(\phi_q^{qe}; \phi_y^r) \equiv \inf \left\{ \phi_{\pi}^r : \max \Re \lambda(D\mathcal{T}^{msv}) < 1 \right\}.$$

This object asks how much additional short-rate response to inflation is needed to offset the destabilizing effect of bond-price-responsive balance-sheet policy within the class of policy rules studied here.

Figure 2 shows how this local policy margin changes with ϕ_q^{qe} . When QE does not react to bond prices, the inflation-response threshold for E-stability lies only modestly above the Taylor-principle benchmark. As ϕ_q^{qe} increases, the threshold rises monotonically. At the benchmark output response $\phi_y^r = 0.125$, it rises from slightly above one to about 1.3. With a larger output response, $\phi_y^r = 0.5$, the threshold increases more sharply. Thus, within this class of rules, a stronger bond-price response

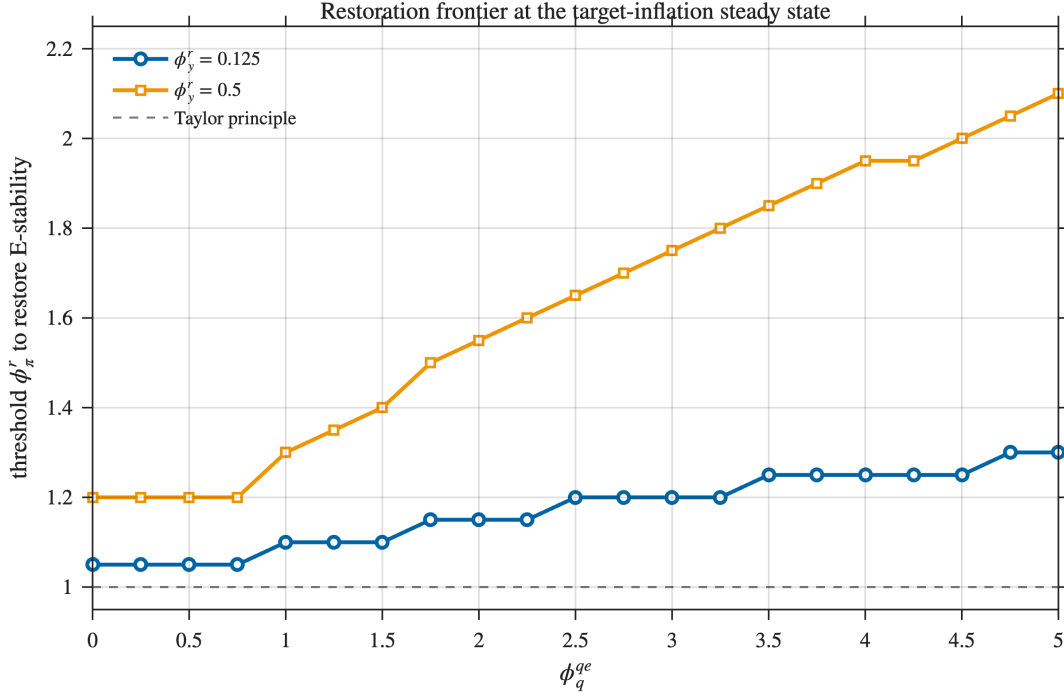


Figure 2: Restoration frontier at the target-inflation steady state. The figure reports the minimum short-rate inflation response $\phi_\pi^{r,*}$ required to restore local E-stability as the QE response to bond prices, ϕ_q^{qe} , increases. The dashed line marks the Taylor-principle threshold $\phi_\pi^r = 1$.

Decreasing-gain adaptive-learning IRFs to a credit shock

$\pi_{\text{ann}}^* = 2\%$, $(\phi_\pi^r, \phi_y^r) = (1.50, 0.125)$, QE on : $\phi_q^{qe} = 3$

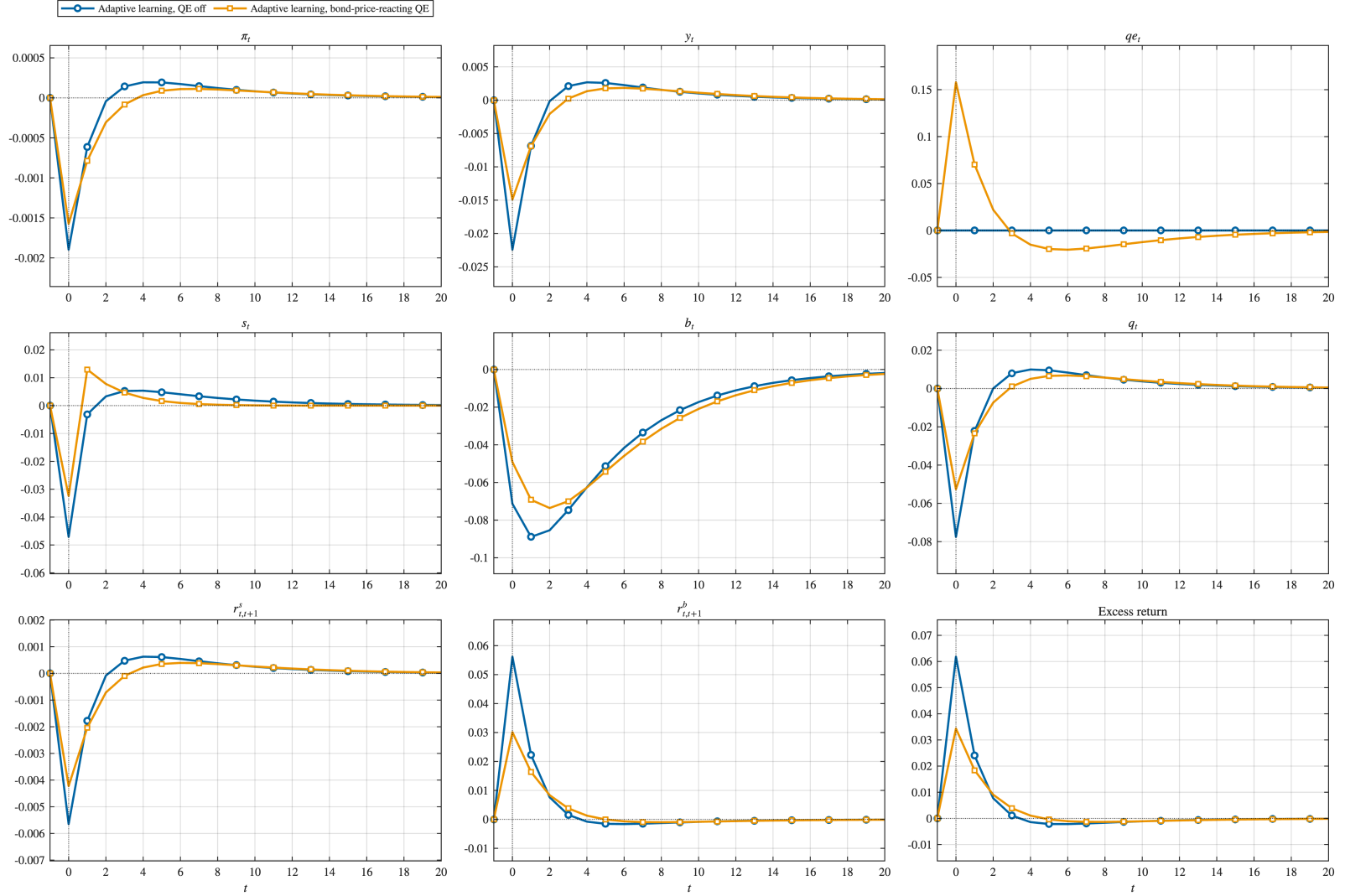


Figure 3: Impulse responses to a negative credit shock under the contemporaneous rule at the target-inflation steady state. The figure compares QE off with bond-price-reacting QE, holding the benchmark short-rate rule fixed at $(\phi_\pi^r, \phi_y^r) = (1.5, 0.125)$.

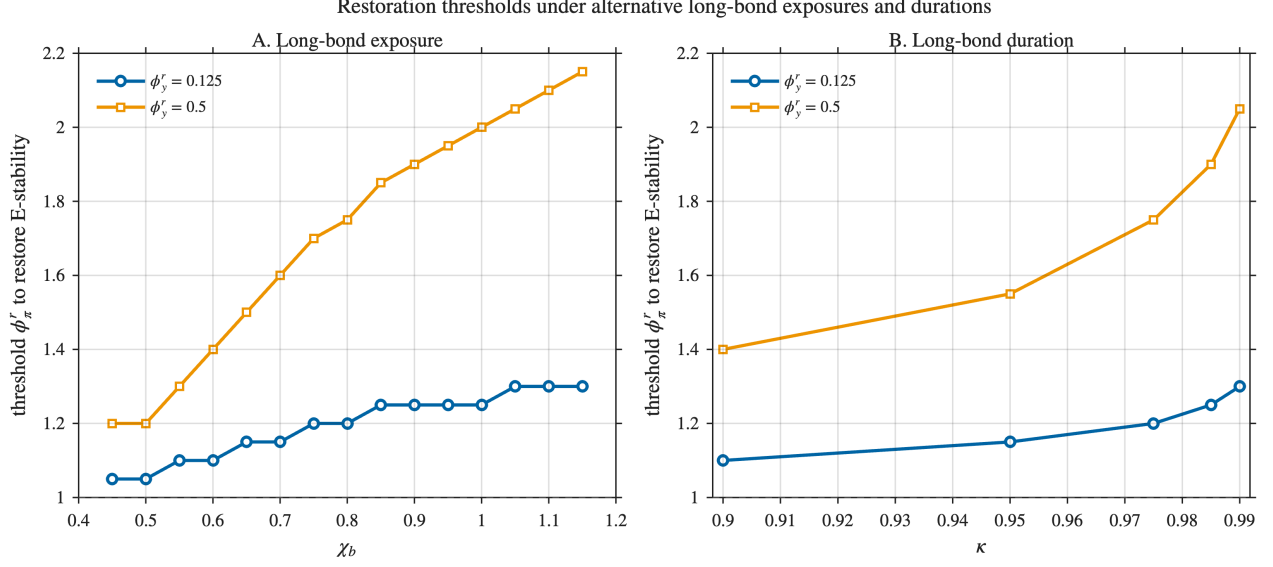


Figure 4: Restoration thresholds under alternative long-bond exposures and durations. The left panel varies the steady-state debt-to-output ratio χ_b . The right panel varies the coupon-decay parameter κ , which governs long-bond duration. The vertical axis reports the minimum ϕ_π^r required to restore local E-stability.

in the balance-sheet rule raises the short-rate inflation response needed to keep the intended equilibrium locally learnable.

Figure 4 shows that the E-stability loss is larger when the long-bond block is **more important**. The left panel varies χ_b , the steady-state long-bond debt-to-output ratio. A larger χ_b raises the inflation-response threshold because it scales the bond-price valuation terms in impatient-household demand and in long-bond-market clearing. The right panel varies κ . A higher κ makes the bond longer lived, strengthens the forward-looking component of bond prices (Equation (25)), and raises the restoration threshold. The destabilizing effect of bond-price-responsive QE is therefore larger when long-bond exposure is higher or when the purchased asset has longer duration.

Figure 5 provides a targeted counterfactual for the direct feedback from bond prices into the QE rule. The object being perturbed is the MSV reduced form from which the T-map is constructed:

$$Z_t = \mathcal{F}_2 \hat{E}_t Z_{t+2} + \mathcal{F}_1 \hat{E}_t Z_{t+1} + \mathcal{F}_w w_t.$$

For any triplet $\mathcal{F} = (\mathcal{F}_2, \mathcal{F}_1, \mathcal{F}_w)$, substituting the PLM-implied forecasts into this

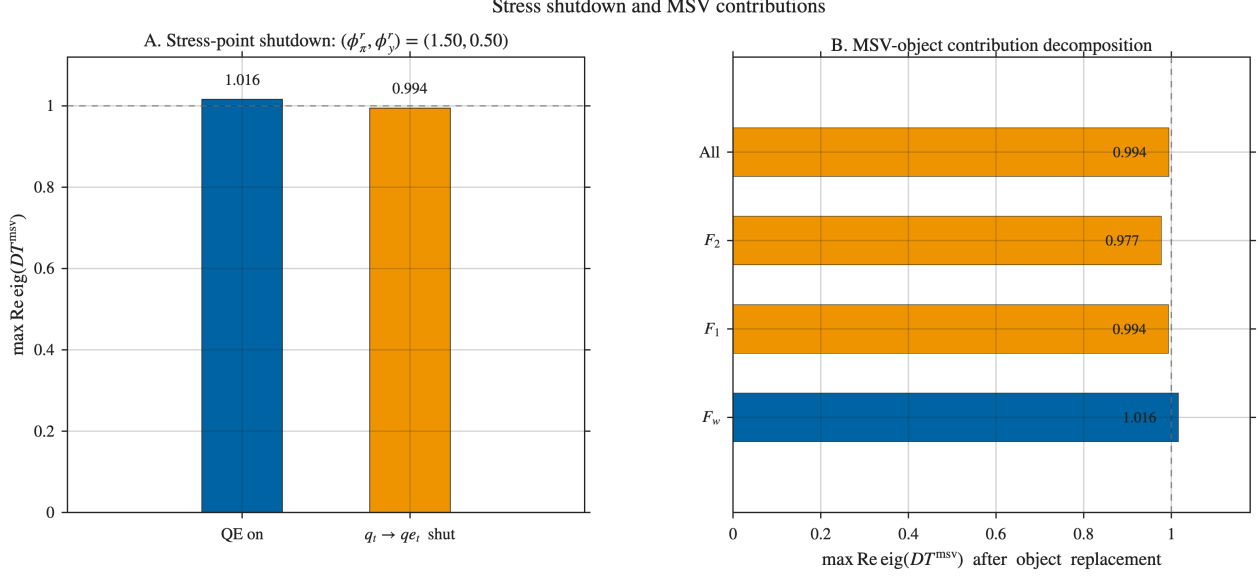


Figure 5: Targeted shutdown of the bond-price-to-QE feedback channel. Panel (a) compares the maximum real part of the T-map Jacobian eigenvalues under the baseline bond-price-responsive QE rule and after structurally shutting down the direct $q_t \rightarrow q_{e_t}$ feedback at $(\phi_\pi^r, \phi_y^r, \phi_q^{qe}) = (1.5, 0.5, 3)$. Panel (b) replaces one MSV reduced-form object at a time by its structural-shutdown counterpart in Equation (41). The dashed line marks the local E-stability threshold.

reduced form gives the MSV T-map $\mathcal{T}_{\mathcal{F}}^{msv}$. The statistic reported in the figure is the maximum real part of the eigenvalues of $D\mathcal{T}_{\mathcal{F}}^{msv}$, evaluated at the corresponding MSV fixed point.

Panel (a) recomputes the MSV fixed point and the Jacobian $D\mathcal{T}^{msv}$ under the full shutdown. The exercise is evaluated at $(\phi_\pi^r, \phi_y^r, \phi_q^{qe}) = (1.5, 0.5, 3)$, matching the policy point used in the active figure-generation code. The shutdown lowers the maximum real part of the T-map Jacobian eigenvalues and moves the system toward the E-stable side of the boundary.

Panel (b) replaces one MSV reduced-form object at a time by its shutdown counterpart and recomputes the corresponding fixed point and $D\mathcal{T}^{msv}$. Replacing either $\mathcal{F}_2$ or $\mathcal{F}_1$ generates most of the stability gain, whereas replacing $\mathcal{F}_w$ leaves the maximum real part much closer to the baseline value. Thus the destabilizing channel operates mainly through the expectational blocks that map bond-price beliefs into the actual law of motion, rather than through the current-state loading alone. The decomposition is not additive, but it shows that the stability loss is concentrated in the

forward-looking part of the MSV learning map.

Taken together, Figure 1–Figure 5 show why the target-inflation results are not contradictory. Bond-price-responsive QE can stabilize simulated transition dynamics because it leans against the financial margin hit by a credit tightening shock. At the same time, by making policy respond directly to q_t , the rule makes the ALM more sensitive to beliefs about the long-bond price, which is forward-looking through the bond-pricing recursion. The shutdown exercise shows that this sensitivity is concentrated in the expectational blocks of the MSV reduced form rather than in the current-state loading alone. Thus the policy can be stabilizing conditional on anchored beliefs while making those beliefs less robust.

V.B. DEFLATIONARY STEADY STATE: THE LOW-RATE EQUILIBRIUM IS NOT A LEARNING ANCHOR

I next turn to the deflationary steady state associated with the ZLB. At the ZLB, the net nominal rate is zero, so the gross policy rate is $\bar{R} = 1$. The steady-state Euler equation then implies $\bar{\Pi}^d = \beta$. Under the benchmark calibration $\beta = 0.995$, quarterly gross inflation at the deflationary steady state is therefore $\bar{\Pi}^d = 0.995$, or about -0.5% per quarter, which annualizes to $(0.995)^4 - 1 \approx -2\%$.

Figure 6 shows that the low-rate equilibrium is not locally stable under learning. With balance-sheet policy inactive, the rate peg $(0, 0)$ lies in the indeterminate and E-unstable region. Allowing asset purchases to react to output or inflation shifts the boundaries only slightly, and the rate peg remains in the same region. When asset purchases react directly to bond prices, the rate peg still is not locally E-stable. Instead, the bond-price rule creates an additional determinate-but-E-unstable wedge for other values of (ϕ_π^r, ϕ_y^r) . Thus balance-sheet policy does not make the deflationary steady state an attractor for nearby beliefs.

This E-instability has a precise local meaning. In a neighborhood of the deflationary steady state, small perturbations to beliefs are not pulled back to the low-rate equilibrium under adaptive learning. The economy therefore need not converge back to the low-inflation steady state after a perturbation. One possible path is toward deeper deflation and weaker activity, which is the sense in which the ZLB equilibrium remains

Contemporaneous rule: deflationary steady state

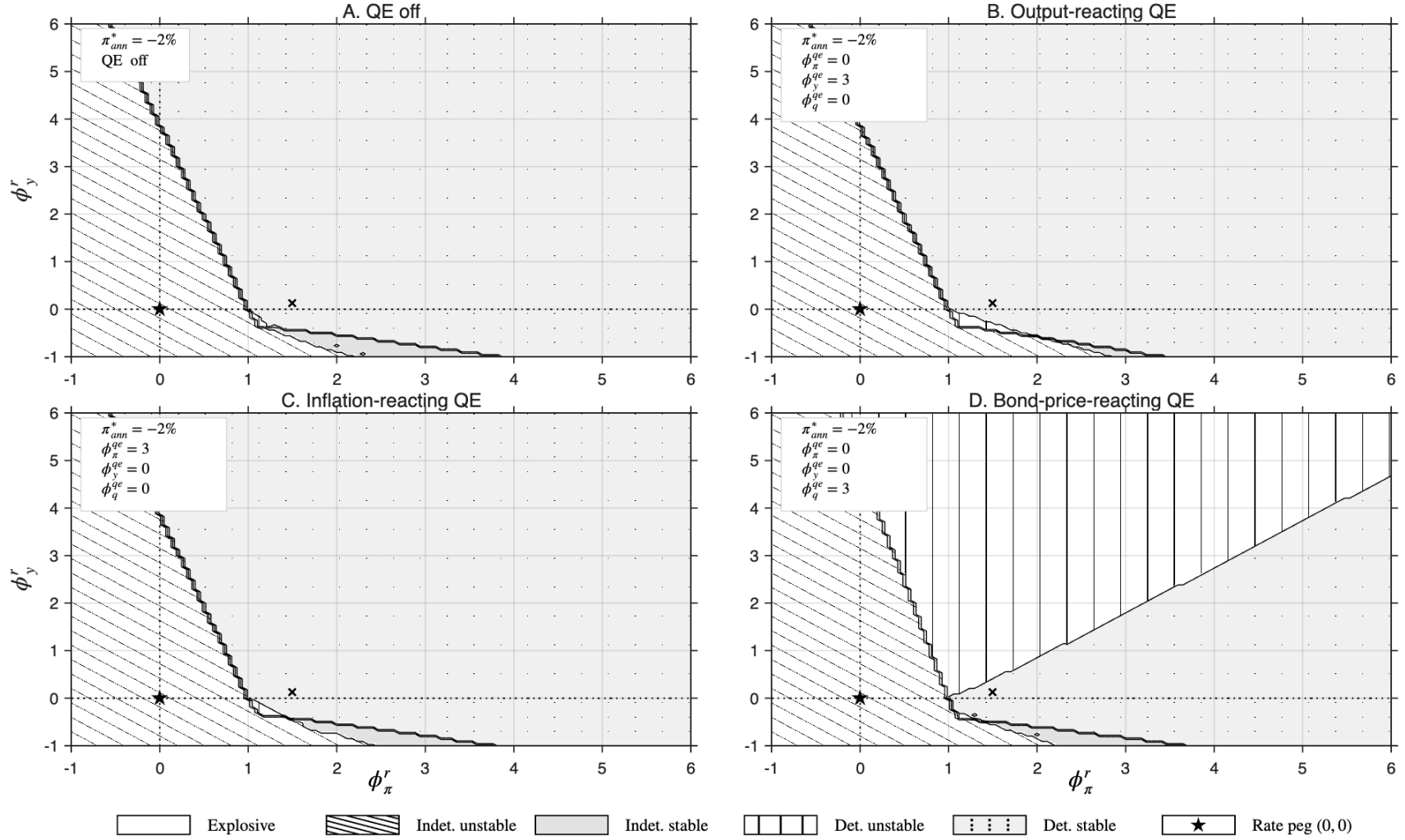


Figure 6: Determinacy and E-stability regions under the contemporaneous rule at the deflationary steady state. The deflationary steady state corresponds to annualized inflation of -2% . The marker “ $\times$ ” indicates the benchmark policy point, and the star marks the rate peg $(0, 0)$, that is, the ZLB policy configuration.

vulnerable to deflation-trap dynamics. But the instability is local, not one-sided: trajectories can also move away from the deflationary steady state and toward the basin of attraction of the target-inflation steady state. The low-rate equilibrium is therefore not a robust anchor for expectations.

VI. CONCLUSION

This paper studies balance-sheet policy when expectations are modeled explicitly rather than imposed through full rational expectations. I extend the framework of Sims et al. (2023) by retaining long-bond prices in the equilibrium system, introducing subjective expectations through Woodford’s (2013) temporary-equilibrium formulation, and replacing rational expectations with adaptive learning in the spirit of Evans and Honkapohja (2001). With trend inflation, the framework permits a unified analysis of determinacy and expectational stability around both the target-inflation steady state and the ZLB deflationary steady state.

Three results emerge. First, QE rules that respond only to inflation or output leave the determinacy frontier, and largely also the E-stability frontier, little changed. The main effect arises when QE responds directly to bond prices: determinacy changes only modestly, but the E-stable region can shrink substantially. Bond-price-reacting QE therefore affects learnability more than rational-expectations uniqueness.

Second, bond-price-reacting QE creates a trade-off between conditional stabilization and expectational robustness. Around the target-inflation steady state, it dampens the response of bond prices, returns, private asset positions, output, and inflation to a credit tightening shock. But the same policy feedback strengthens the link from beliefs about bond prices to realized outcomes, making the intended equilibrium less robust under adaptive learning. Within the class of rules studied here, restoring local learnability then calls for a stronger short-rate inflation response. This local requirement is amplified when long-bond exposure is larger or when the purchased asset has longer duration.

Third, balance-sheet policy does not make the ZLB deflationary steady state a learning anchor. At that steady state, the rate peg remains in the indeterminate and E-unstable region, and allowing asset purchases to react to output, inflation, or bond

prices does not move the economy into a learnable low-rate equilibrium. This local instability should not be interpreted as necessarily undesirable. Belief perturbations are not pulled back to the deflationary steady state, but they may move either toward deeper deflation or away from the low-rate equilibrium toward the target-inflation basin.

Taken together, these results imply that QE should not be evaluated solely by its effects on equilibrium allocations under rational expectations. Because QE operates through expectations-sensitive financial variables, its performance also depends on whether private beliefs converge to the intended equilibrium. A rule that appears benign under rational expectations may therefore be much less robust once learnability is imposed as an additional criterion. For unconventional monetary policy, the relevant issue is not only transmission, but also expectational coordination.

The policy lesson is therefore conditional rather than absolute. The results do not imply that balance-sheet interventions are ineffective. They imply that the feedback variable in the asset-purchase rule matters for learnability. Rules that mechanically react to forward-looking asset prices can weaken expectational robustness even when they stabilize responses to financial shocks. In the local environments studied here, robustness is better preserved when the short-rate rule anchors inflation expectations and the balance-sheet rule is not made directly responsive to bond-price movements.

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